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Soft electrodes for simultaneous bio-potential and bio-impedance study of the face

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The human body's vascular system is a finely regulated network: blood vessels can change in shape (i.e. constrict, or dilate), their elastic response may shift and they may undergo temporary and partial blockages due to pressure applied by skeletal muscles in their immediate vicinity. Simultaneous measurement of muscle activation and the corresponding changes in vessel diameter, in particular at anatomical regions such as the face, is challenging, and how muscle activation constricts blood vessels has been experimentally largely overlooked. Here we report on a new electronic skin technology for facial investigations to address this challenge. The technology consists of screen-printed dry carbon electrodes on soft polyurethane substrate. Two dry electrode arrays were placed on the face: One array for bio-potential measurements to capture muscle activity and a second array for bio-impedance. For the bio-potential signals, independent component analysis (ICA) was used to differentiate different muscle activations. Four-contact bio-impedance measurements were used to extract changes (related to artery volume change), as well as beats per minute (BPM). We performed concurrent bio-potential and bio-impedance measurements in the face. From the simultaneous measurements we successfully captured fluctuations in the superficial temporal artery diameter in response to facial muscle activity, which ultimately changes blood flow. The observed changes in the face, following muscle activation, were consistent with measurements in the forearm and were found to be notably more intricate. Both at the arm and the face, a clear increase in the baseline impedance was recorded during muscle activation (artery narrowing), while the impedance changes signifying the pulse had a clear repetitive trend only at the forearm. These results reveal the direct connection between muscle activation and the blood vessels in their vicinity and start to unveil the complex mechanisms through which facial muscles might modulate blood flow and possibly affect human physiology.

1. Introduction

The control of blood flow in the face holds significance in facial physiology, encompassing aspects like facial expressions and skin well-being [1, 2]. Blood circulation is subject to a range of factors, including the expansion and narrowing of blood vessels. Muscle contraction in the proximity of blood vessels can cause them to narrow, and thus blood circulation in the face can potentially be locally influenced by facial muscles [1, 3]. Facial muscles, compared to other skeletal

muscles of the body, are unique: they are connected to the skin (rather than being fixed to bone-based insertion points), they are interwoven and they partially overlap each other [4, 5]. Consequently, facial movements have complex activation patterns, where nearly all facial muscles are co-active [6]. Scientific explorations spanning over decades explored many phenomena regarding the physiological, psychological, and cognitive effects of facial muscle activation. For example, expansive evidence implicates facial muscle activation with the expression of emotions

[7–10]. Moreover, facial muscles are associated with a wide range of medical disorders including Parkinson's disease (hypomimia, the reduction or loss of spontaneous facial expressions) [11], Tourette syndrome (fast, brief, and repetitive facial movements in the form of tics) [12] and chronic facial palsy (might lead to synkinetic muscle activity or muscle atrophy) [13, 14].

Anatomically, facial muscles and the vascular network are interwoven [15, 16], suggesting a mechanical mechanism in which dynamic changes in facial muscle activation may affect blood flow. It was long ago empirically suggested that facial muscles play a role in modulating blood flow [17, 18]. Recently, several approaches were used to explore the dynamic interaction between blood flow and facial muscles. Ultrasound was used to study blood flow blockage caused by muscle activation during smiling [1]. The interaction between facial muscles and blood flow was also recorded in photoplethysmography (PPG) measurements in the ear during facial expressions [3]. In parallel, attempts were made to show an indirect effect of this interaction. Mainly, if facial expressions have an effect on blood flow or temperature in the brain [19]. For example, using functional magnetic resonance imaging (fMRI), lower brain activity (in the amygdala) was reported during a mimicry of angry expressions when the forehead's muscles were paralyzed [20]. Similarly, an emotional state from spontaneous expressions was assessed using functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) [21]. However, such attempts do not offer insight into local mechanical muscle-vessel interactions.

Overall, the study of the link between facial muscles and a change in blood flow due to changes in local vessel diameter was so far limited, with existing tools allowing only low-resolution investigation under static conditions. To facilitate the investigation of the interaction between facial muscles and blood flow in the face under natural conditions, we introduce a novel skin electronic approach, consisting of impedance plethysmography (IPG), combined with a high-resolution surface electromyography (sEMG). IPG, sEMG, and EEG from the face under natural conditions were achieved with printed soft electrode arrays. Using this method, we recorded high-resolution facial muscle activation and simultaneous minute changes in IPG at the superficial temporal artery to investigate the impact of facial muscle activation on vessel diameter, which potentially affects blood flow. We further explore how this relation differs at the forearm.

2. Methods

2.1. Subjects

Four healthy volunteers (ages 27–30) participated in the study. For both the sEMG and IPG measurements, an electrode array was adhered to the skin. All

experiments were conducted in accordance with relevant guidelines and regulations under approval from the Institutional Ethics Committee Review Board at Tel Aviv University. Informed consent was obtained from all subjects. The research was conducted in accordance with the principles embodied in the Declaration of Helsinki and in accordance with local statutory requirements.

2.2. Soft IPG and sEMG electrodes

The electrode arrays used in this study are based on a screen-printing process on soft substrates, as previously described [22, 23]. Briefly, carbon electrodes and silver traces were screen-printed on a thin and soft polyurethane (PU) film (80 μm from Delstar EU94DS). A second double-sided adhesive PU film was used for passivation and skin adhesive material. The sEMG arrays (X-trodes Inc.) consisted of 16 electrodes with an additional internal ground.

2.3. sEMG measurement and analysis

sEMG data were recorded with a miniature wireless data acquisition unit (DAU, X-trodes Inc.). The DAU supports up to 16 unipolar channels (2 μV noise RMS, 0.5–700 Hz) with a sampling rate of 4000 S/s, 16 bit resolution and input range of ± 12.5 mV. A 620 mAh battery supports DAU operation for a duration of up to 16 hr. A Bluetooth (BT) module is used for continuous data transfer. The DAU is controlled by an Android application and the data are stored on a built-in SD card and the Cloud for further analysis. The DAU also includes a 3-axis inertial sensor, to measure the acceleration of the hand or face during the measurements.

The face electrode array was placed at close estimated proximity to major facial muscles, such as the zygomaticus, frontalis, and orbicularis oris muscle. The forearm electrode array was positioned at the flexor carpi radialis and pronator teres muscles region. A 16-electrode array was placed such it is centered approximately at one-third of the forearm anterior length measured from the wrist to the elbow (see figure 1). The array was connected to a wireless DAU.

ICA was performed following the scheme presented previously [22, 23]. Here we implemented the algorithm in Python, using python-picard [24, 25]. We also implemented the heat maps described previously to visually present the sources on the face, allowing us to discern the different segments of muscles that were activated. The interpolation was done using Scipy (griddata, linear method). In addition, to evaluate the dominance of each ICA source, we calculated their intensities in each action: First, the absolute value of the amplitude of each source during each action was calculated, then the intensity was defined as the average of the absolute value.

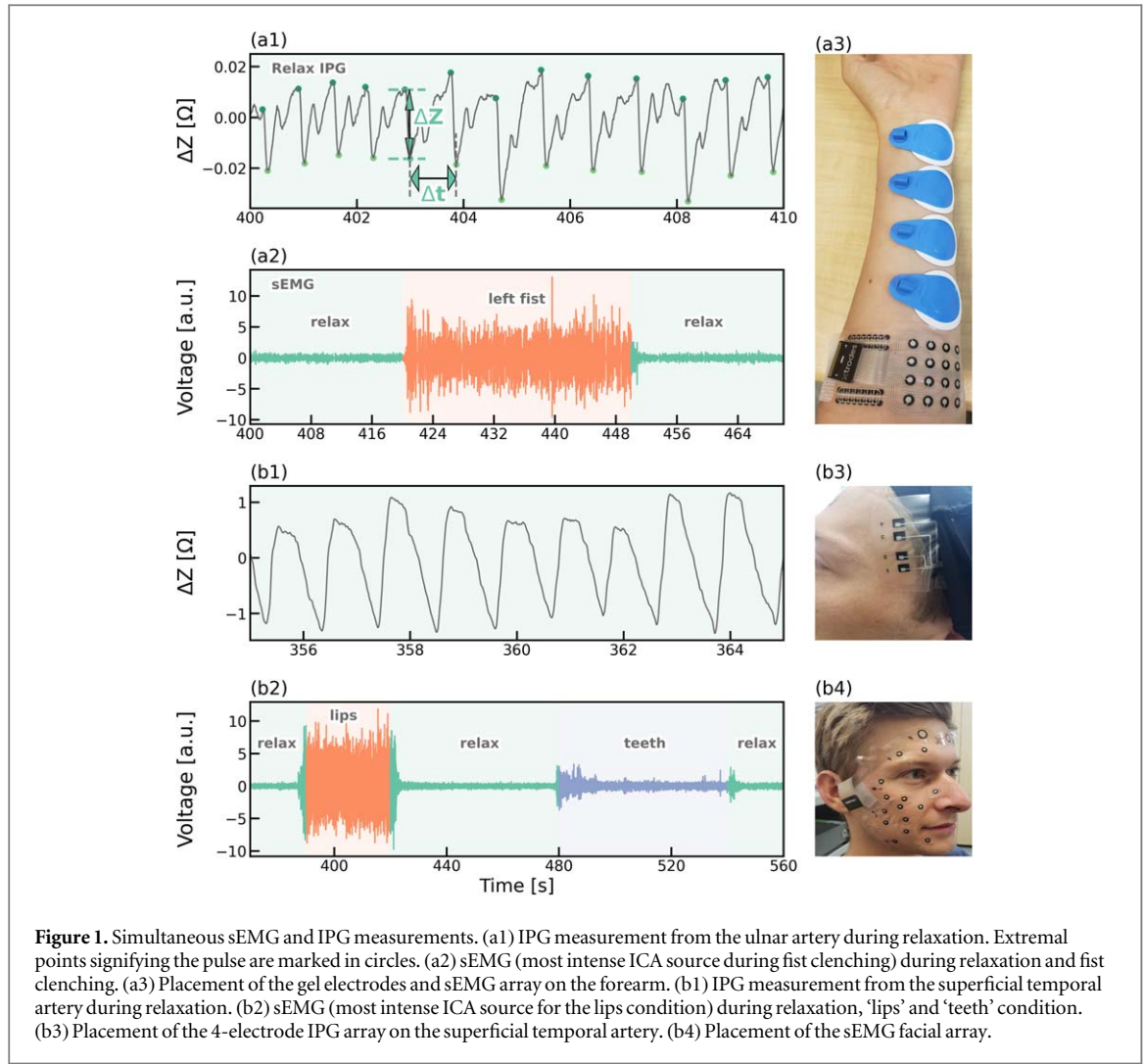


Figure 1. Simultaneous sEMG and IPG measurements. (a1) IPG measurement from the ulnar artery during relaxation. Extremal points signifying the pulse are marked in circles. (a2) sEMG (most intense ICA source during fist clenching) during relaxation and fist clenching. (a3) Placement of the gel electrodes and sEMG array on the forearm. (b1) IPG measurement from the superficial temporal artery during relaxation. (b2) sEMG (most intense ICA source for the lips condition) during relaxation, 'lips' and 'teeth' condition. (b3) Placement of the 4-electrode IPG array on the superficial temporal artery. (b4) Placement of the sEMG facial array.

2.4. IPG measurement and analysis

The IPG array consisted of four electrodes to support a 4-terminal measurement. The four-contact IPG was measured using an impedance analyzer (MFIA by Zurich Instruments). A custom-made printed circuit board (PCB) was produced to facilitate contact between the MFIA and the soft electrodes which were connected directly to flat conducting traces on the PCB using a z-axis adhesive. IPG data were analyzed with Python code. First, a running average over the raw data was performed, using Scipy (uniform_filter1d). Then, the detection of the maximum and minimum peaks of the pulse was performed (Scipy, find_peaks). Next, the impedance change and beats per minute (BPM) were calculated as follows: The impedance change ΔZ was defined as the difference between the absolute impedance at the maximum and the absolute impedance at the minimum of each pulse (see figure 1). As pulsation measurements included dynamic actions, only reliable pulses were kept. Reliable pulses were defined as those with a repetitive pulse shape and behavior, with approximately equal spacing. For the BPM, we calculated the time between each peak minima (each pulse), Δt , then the BPM was

defined as one over Δt , multiplied by 60 (as Δt is in seconds). See figure 1.

The impedance can be related to the diameter of the blood vessel by modeling the artery as a cylindrical resistor [26]:

$$Z_a = \rho l / A = \rho l / (\pi r_a^2) \quad (1)$$

where Z_a is the impedance of the artery, r_a is the artery radius, l is length of the measured segment, and ρ is the blood resistivity.

The overall bio-impedance is also sensitive to blood pulsation and can be therefore described by a baseline impedance (Z) and ΔZ - the change in the impedance due to pulsation. ΔZ can be expressed by equation (2) [27]:

$$\Delta Z = \pi Z^2 \Delta r_a^2 / (\rho l) \quad (2)$$

where $\Delta r_a = \sqrt{r_a'^2 - r_a^2}$ is the (artery) radius pulsation amplitude change.

Further detail can be found in Supplementary Material E.

2.5. Fist clenching protocol

A 16 electrode sEMG array was placed on the forearm, together with four commercial pre-gelled Ambu

electrodes of 40 mm in diameter (Ambu® BlueSensor Q ECG electrodes) along the ulnar artery. Subjects were instructed to rest and then to clench their fist. Muscle activity (sEMG) and blood pulsation (IPG) were recorded during fist clenching and release.

2.6. Facial muscle activation protocol

Subjects with sEMG and IPG electrodes were recorded while performing three facial activation tasks: ‘lips’ (holding a pen in the mouth with the lips), ‘teeth’ (holding a pen in the mouth with the teeth), and facial expressions. The ‘lips’ and ‘teeth’ tasks were adapted from Strack and coworkers [28]. We chose this task as it is a simple mechanical facial ‘expression’ that can be activated for longer without feeling overly self-conscious. We also tested sEMG and IPG during facial expressions. IPG measurements were taken from the superficial temporal artery, as it is easily accessible with minimal discomfort to the subject.

3. Results

Soft and dry electrodes were used for simultaneous IPG and sEMG measurements, to study how muscle activation affects the diameter of the superficial temporal artery in the face. We used unique screen-printed dry carbon electrodes which were extensively validated previously for bio-potential [29] and bio-impedance [30] measurements. As a reference, we also performed similar measurements on the forearm. In both cases, sEMG data were recorded with soft 16 electrode arrays and a miniaturized and wireless DAU. IPG was measured using a 4-terminal impedance measurement. For IPG from the forearm, both dry and conventional gelled electrodes were used (see supplementary figure B1-B2). Clearly, to access arteries in the face, IPG measurements had to be performed using low form factor and soft electrodes that can conform to the curvature of the face.

In figure 1, we show the two electrode arrays (i.e. IPG and sEMG) placed on the forearm and the face (figure 1(a3)) and IPG electrodes along the ulnar artery and on the face (figure 1(b3)). The advantage of the dry-printed electrodes to study the face is readily apparent.

In figure 1 we also show typical IPG and sEMG data from the arm and the face. For the sEMG data, we used independent component analysis (ICA), to minimize noise and to reduce the effect of cross-talk [23]. For clarity, the onset of muscle activity is marked by orange in figure 1(a2) for the forearm and orange and blue in figure 1(b2) for the face. IPG data was recorded simultaneously. As clearly seen in figure 1(a1) and 1(a2), IPG records blood pulsation that can be readily seen when the electrodes are positioned in close proximity to an artery (ulnar artery in the forearm or superficial temporal artery in the face). As we recently demonstrated in [30], IPG recordings with soft-

printed dry electrodes are characterized by high-quality signals and good stability, even under muscle activation.

IPG data can be used to derive artery properties. Foremost, we observe that the pulse shape differs between the forearm and face, with the secondary notch (the dicrotic notch [31]) being less pronounced at the face. Indeed pulse shape has a strong association with the electrode location [30, 32]. Additionally, from the extremal points (corresponding to the pulsation itself), we can extract the pulse amplitude (ΔZ) and duration (Δt) (see figure 1(a1)), which characterize blood flow. Pulsation amplitude provides information about the artery mechanical parameters (equation E.3), and the heart rate. Moreover, baseline impedance can be associated with artery cross-section (equation E.1-E.2). Simultaneous IPG and sEMG measurements reveal how muscle activation impacts artery properties at rest and during muscle action.

In figure 2, we examine how fist clenching affected the diameter of the ulnar artery. Figure 2 shows sEMG activity at the forearm muscle of the left hand during five repetitions of fist clenching and relaxation. The IPG signal is directly affected by muscle activation in a very consistent manner (figure 2(a)). At the onset of muscle activation, the baseline IPG signal shifted upward (i.e. the impedance Z increases, the artery diameter decreases, figure 2(a) bottom panel) and the pulse amplitude ΔZ slightly increased (figure 2(b) top panel). At the release of the force, the IPG signal returned to the baseline (artery diameter returns to its original state). After the release of the force, we also observed a change in IPG pulsation amplitude. It first deflected downwards and gradually returned back to the baseline value (figure 2(b) middle panel). Finally, we make note of the elevated heart rate during fist clenching (from 75 ± 5.6 BPM at initial relaxation to an average of 92 ± 6.8 BPM across all fist clenching repetitions, figure 2(b) bottom panel).

Under manual obstruction of the artery (both above the elbow, far from the measurement site (Supplementary figure (A1)-(A3)), and at the forearm at close proximity to the measurement site), we observed similar behavior in Z (increase during blockage and return to the baseline after release). In both fist clenching and mechanical blockage, the increase in Z can be associated with artery radius reduction, followed by a return to the baseline when the blockage is lifted. ΔZ was slightly larger during fist clenching than during relaxation (figure 2). As explained in the Supplementary, how Δr_a changes reflects on the viscoelasticity of the artery and may depend on the manner the blockage is applied and released. The measurements were repeated both with gel electrodes and printed carbon electrodes and similar results were obtained (Supplementary figure (B1)-(B2)).

For the study of facial muscles, we followed the approach of Strack and co-authors [28] to generate different activations. Two facial actions were used:

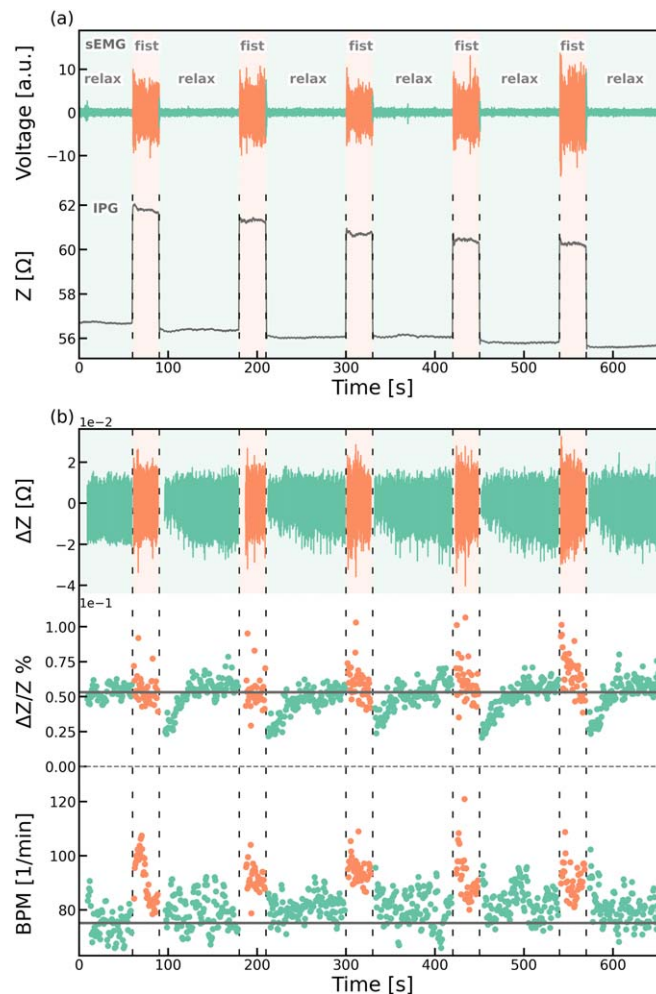


Figure 2. sEMG and IPG response to fist clenching. (a). Most intense ICA source during fist clenching (top) and 4-terminal IPG signal at the ulnar artery as a function of time (bottom). (b). Running average of the impedance as a function of time (top), relative impedance changes as a function of time (middle), BPM as a function of time (bottom). Vertical dashed lines signify a transition between states and the solid black line signifies the average at the first relaxation period. Only data from detectable pulses are shown.

'lips' and 'teeth'. In 'lips', the participants were asked to hold a pen with their lips, while in 'teeth', a pen was held with their teeth. In this extensively studied paradigm, the 'lips' action supposedly inhibits smiling, while the 'teeth' action activates muscles necessary for smiling [28]. In fact, in our high-resolution facial sEMG data (data discussed below), we can map which muscles are active, and show how these activations change from action to action.

As portrayed in figure 3, simultaneous IPG and sEMG measurements from the face reveal a conspicuous link between muscle activation and vessel diameter. Foremost, at the onset of muscle activation, the baseline IPG signal shifts upward (increase in impedance, decrease in artery radius), as shown in the bottom panel of figure 3(a). The increase or decrease of pulse amplitude, both during and after muscle activation, differed between the two actions. In figure 3(b) (top and middle panels) we show an example where the pulsation amplitude increased slightly during muscle activation in the 'lips' condition and increased more substantially in the 'teeth' condition (see other examples in Supplementary Material C). Interestingly,

in contrast to the effect observed during fist clenching, after the force is released, the pulsation amplitude does not exhibit an abrupt decrease with a gradual increase to the baseline (across all repetitions). In general, and as shown here (in figure 3(b), top and middle panels), after muscle activation, pulse amplitude increased. The qualitative difference between the ulnar and the superficial temporal artery may be attributed to their different arterial properties, taking into account that the determining factors for arterial stiffness stem from the cellular and extracellular composition of the artery, as well as its geometry [33, 34].

Altogether these results show that the difference between the two movements (smile and non-smile) is manifested in different muscle activations (amplitude and composition) as well as changes in vessel diameter. The implications of these results are further discussed below.

Unlike the robust and consistent behavior observed during fist clenching, the effect observed in the face is more complicated, potentially reflecting the higher complexity of the facial muscle-blood network and the intricate manner in which arteries are

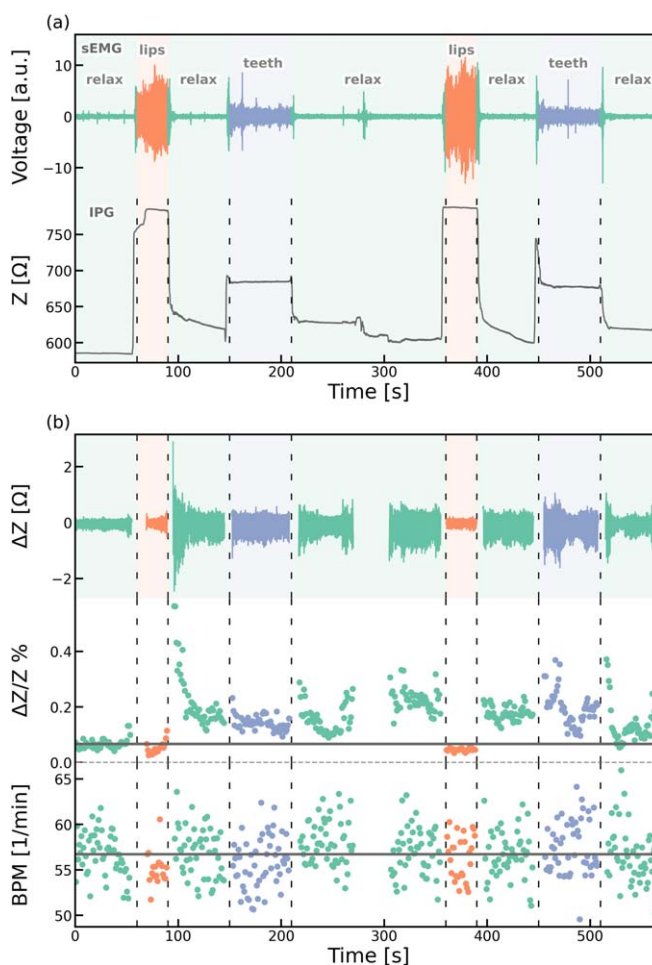


Figure 3. sEMG and IPG response to Strack conditions ('lips' and 'teeth'). (a). sEMG activity versus time (most intense ICA source during the 'lips' condition) and 4-terminal IPG signal at the superficial temporal artery versus time (bottom). (b). Running average of the impedance versus time (top), relative impedance change versus time (middle), and beats per minute (BPM) versus time (bottom). Dashed lines signify a transition between states and the solid black line is the average at the first relaxation period.

interwoven between the muscles [2]. Repetition of the protocol showed different responses in the pulsation amplitude. We hypothesize that the cause for the variability in the IPG responses (in the face) is due to differences in muscle activation, even when similar action is performed. We explore this in figure 4. Figure 4 demonstrates the distinct difference between the forearm and the face. We plot all ICA sources during each activation, ordered by intensity (highest to lowest, left to right), the color denotes the average intensity of the source throughout all sessions. The case of fist clenching is typified by close similarity between repetitions. This result is echoed in the IPG data: for all repetitions ΔZ during fist clenching increases between 2.6–5.8 standard deviations (σ) from ΔZ_0 (changes during initial relax). Similarly, post fist clenching, ΔZ decreases between 0.72–1.9 σ from ΔZ_0 . In the face, each activation is different. We can see how the contribution of the source varies in different sessions, despite the fact that the measurements were taken from the same participant, on the same day, using the same electrode arrays and on the same location. In particular, source 6 (forehead) was

weak in the first sessions, but is one of the top most intense sources in the last three lips/teeth activations. Similarly, source 10 (corner of the lips), was the strongest source for both the 'lips' and 'teeth' condition in the first and second sessions, drops to the second and third least intense sources for the third and fourth sessions. This is in contrast to the muscle activation in the forearm, where variability of the muscles activated is much smaller (figure 4(a)). Indeed, the variability of muscle activation between each fist clenching is on average 0.58, meaning on average the intensity of the sources varies up to 1 ranking. The average variation for the face, on the other hand, is 1.6 for the 'lips' and 2 for the 'teeth' condition. This means the ranking of the intensity varies on average by 2 rankings in the face, twice as much as in the forearm (Supplementary figure D(1)). This variability is particularly manifested in ΔZ for the teeth condition, as well as the post-action relaxation period in both 'teeth' and 'lips'. For the former, ΔZ differs from ΔZ_0 anywhere from 2.4 to 8.8 σ , and the latter has a wider range still of 1.5–24 σ .

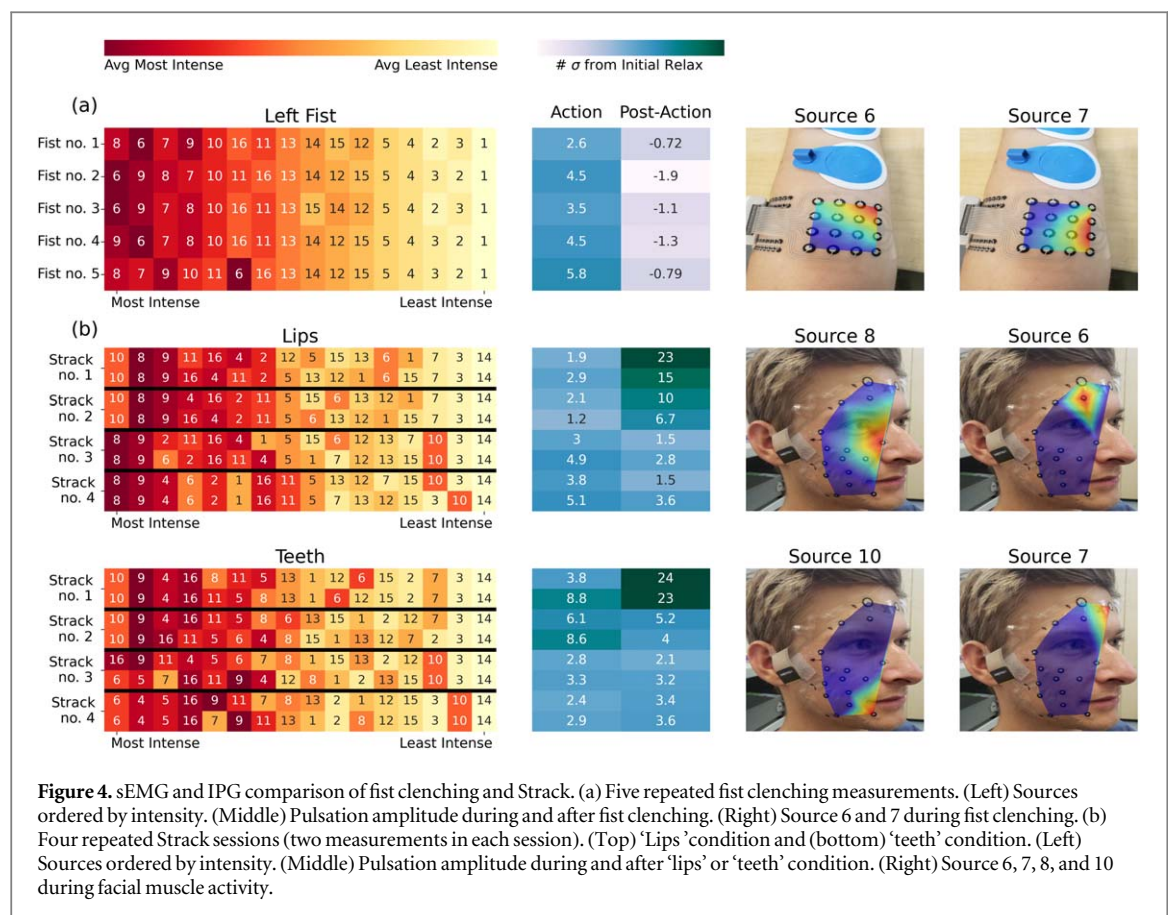


Figure 4. sEMG and IPG comparison of fist clenching and Strack. (a) Five repeated fist clenching measurements. (Left) Sources ordered by intensity. (Middle) Pulsation amplitude during and after fist clenching. (Right) Source 6 and 7 during fist clenching. (b) Four repeated Strack sessions (two measurements in each session). (Top) 'Lips' condition and (bottom) 'teeth' condition. (Left) Sources ordered by intensity. (Middle) Pulsation amplitude during and after 'lips' or 'teeth' condition. (Right) Source 6, 7, 8, and 10 during facial muscle activity.

4. Discussion

In this paper, we discussed a novel electronic skin technology designed for analyzing facial characteristics. Through the utilization of soft and non-invasive electrode arrays, we achieved the capability to simultaneously record both bio-potential and bio-impedance data. This allowed us to effectively record delicate variations in the diameter of the superficial temporal artery as a response to facial muscle movements.

We report on direct evidence for a mechanical muscle-vessel connection. Both in the forearm and the face, simultaneous sEMG and IPG data reveal a clear change in the baseline impedance (Z) during muscle action. The baseline impedance increases, indicating blood vessel narrowing.

A second phenomenon we investigated in this paper is the effect of muscle activation on pulse amplitude. During fist clenching, the observed impedance change due to pulsation (ΔZ) consistently increased under repetitive actions. In the face, the increase (if occurs) varied from activation to activation, in particular for the 'teeth' condition.

Finally, we examined the post-action impedance change. Unlike the consistent effect in the forearm, in the face, we observed a change that depends on the exact muscle action as revealed by the high-resolution sEMG analysis. In the forearm, the post-action impedance change ΔZ decreased with a stable recovery to the baseline. No such trend was observed in facial

muscle activation. It is interesting to see that the impedance change (ΔZ) post-action is comparable, if not even greater, to the impedance change in the 'lips' or 'teeth' condition, despite the fact that muscle activity is stronger during the actions. The variability in this effect emphasizes the importance of what combination of muscles is activated over their magnitude. It demonstrates the complexity of the facial muscle-vessel network: In the face, seemingly similar expressions could activate different combinations of muscles, which could lead to a different outcome. Each combination of activated muscles could result in a different effect on the IPG signal.

Thus, an important result of our investigation is how the 'lips' and 'teeth' actions differ in their IPG response: Both lower amplitude and faster recovery are observed in 'teeth' versus 'lips'. Clear responses of the IPG signals to spontaneous facial expressions (Supplementary figure C10-C15) and to yawning (Supplementary figure C8-C9) suggest an underlying mechanical mechanism. A more expansive investigation is needed to corroborate these findings.

The experimental approach we discussed here allowed us to observe the direct mechanical link between muscles and blood vessels, both in the forearm and in the face. Increased blood supply to active muscles is the most studied muscle-blood phenomenon [35–37]. Direct mechanical muscle-blood connection is a documented mechanism in which contracting skeletal muscles squeeze vasculature,

against underlying tissue or bones, causing an effective narrowing of these blood vessels. The most well-known example of this regulatory interaction is the muscular pump in the leg [38–40]. Evidence for a mechanical muscle-blood connection in the forearm is also reported [41]. And here, for the first time, we mapped these effects in the face.

Human blood circulation is complex. It is a carefully regulated system in which chemical, neural, and mechanical processes take place in concert to reach proper flow through the network of arteries and veins in response to changing physiological demands. Abnormalities such as changes in wall rigidity, mechanical constrictions or long-term application of pressure change the properties of vessels and indicate potentially fatal medical conditions such as stroke and cardiovascular diseases [42–44]. Thus, the reaction of vessels to the mechanical pressure of muscles can serve as a prognostic tool, and perhaps early detection. The results we discussed here reveal the complex manner by which facial muscles can influence blood flow through changing vessel diameter in the face, shedding new light on how facial motion can affect physiology, and perhaps even our psychological state [8, 17, 18, 28]. Future studies looking into the link between facial expressions and emotion might benefit from inspecting both muscle activity and blood flow. A possible additional significance of the action of facial muscles on blood vessels is in assessing clinical outcomes of facial surgeries [43–45]. Lastly, the results point out a possible route towards expanding in-vivo analysis of bio-mechanical properties of arteries [1], for example, investigation of the active response of arteries [46].

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Data availability statement

The datasets generated and/or analyzed during the current study are not publicly available as they rely on face images. The data that support the findings of this study are available upon reasonable request from the authors.

Conflict of interest

YH declares a financial interest in X-trodes Ltd, which developed the screen-printed electrode technology used in this paper. YH has no other relevant financial

involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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